

Department Technology Systems

Health Information (HI) professionals use technology to facilitate the ease of entering, analyzing, using, and maintaining data. Advances in the types and capabilities of information systems available to HI professionals have significantly impacted the way health information and related data are shared and communicated. What once were laborious and repetitive tasks can now be accomplished fairly simply without the need to duplicate work, thus increasing productivity and decreasing the possibility of errors.

Common Software Applications

There are some systems that are for use only in the HI department such as an encoder for coding. Other systems, such as the master patient index (MPI), are used in collaboration with other departments throughout the whole organization. Systems that support the HI department include release of information systems to encoders and chart deficiency systems to transcription systems. An understanding of these systems contributes to the efficiency of overall operations within the department.

HI professionals also use many types of software applications in their daily work, including the following:

Words

For HI professionals, this software is often used in creating reports, performance evaluations, patient and meeting records, and employment advertisements and materials. Across all specialties, word processing is central to the daily reporting and patient management tasks necessary to efficient healthcare.

HI professionals use spreadsheets to track and report inventory, personnel, billing, and operating budgets. When the time comes to report important statistics to regulatory agencies, spreadsheets will assist to gather large sets of data to be transferred to the necessary departments. HI professionals play a vital role, as they are the main hub of data collection and sharing.

Databases are useful ways to organize large amounts of information. Information systems store and run programs to easily access the data and perform calculations. These same databases can be accessed by

many healthcare professionals including doctors, lab technicians, and research scientists. Wherever there is statistical data to be gathered, databases provide a powerful way of storing that data in a small amount of physical space and making it accessible to others.

Encoder Systems

Providers must document a patient's condition and/or diagnoses and all procedures performed during each encounter in the health record (paper or electronic). The healthcare organization is then responsible to assign codes to the diagnosis and procedures by following the applicable coding classification/code set conventions and guidelines. Diagnosis codes are assigned using the International Classification of Diseases, Tenth Revision, Clinical Modification (ICD-10-CM) and procedures are assigned using the International Classification of Diseases, Tenth Revision, Procedure Classification System (ICD-10-PCS) for inpatient procedures or Current Procedural Terminology (CPT®) for outpatient procedures.

An **encoder** is a computer software program that is used to assign the codes instead of using the hard copy code books. Many encoders also include grouper software that uses the codes selected to calculate the healthcare organization's reimbursement for the patient encounter. Encoders and groupers are used primarily within the HI department. Therefore, the HI professional will generally lead the group that selects, implements, does the training and maintains the system.

The 3M™ Coding and Reimbursement System is a suite of tools used to generate accurate, complete, and compliant grouping for patient records. Access the 3M™ Coding website ([Health Information Systems](#)) and follow the steps below to become familiar with this coding and reimbursement system.

Step 1

Within the menu item *Improve Revenue Cycle*, select and review each of the following solutions/services:

- Clinical documentation integrity
- Coding
- Patient classification methodologies

Step 2

review information about the following available systems:

- 3M™ Coding Reference Plus Software
- 3M™ Advanced Analyzer Software
- 3M™ APCfinder Software

- 3M™ APR DRG Software
- 3M™ Codefinder™ Software
- 3M™ Coding Reference Software

Automated Coding Workflow

"The coding workflow process varies by provider setting and other organizational factors such as the amount of noncoding activities performed by coding professionals. This lack of a standardized work process requires each organization to define the current and future state of coding and how CAC fits into the process for themselves."

Release of Information Systems

To ensure that each stop in the patient's healthcare journey has all the information needed to give consistent and quality treatment, it's vital that the patient's health records are accessible when needed. HI is the connection between the patient and clinicians that ensures the information needed is disclosed per regulatory guidelines to protect the patient's privacy but not hinder them from receiving timely, appropriate, and quality treatment. Release of Information (ROI) systems facilitate this communication.

Management Practices for the Release of Information

"Exchange of health information is an essential function for the provision of high-quality, cost-effective, and safe healthcare. The health information released must be complete, appropriate, and timely to fulfill its intended purpose. Although this sounds straightforward, often it is not an easy task in the complex medical and legal environment in which the healthcare community operates. Release of information (ROI) in healthcare is critical to the quality of the continuity of care provided to the patient. It also plays an important role in billing, reporting, research, and other functions."

Cancer Registries

Data collected by state cancer registries enable public health professionals to understand and address the cancer burden more effectively. The Centers for Disease Control and Prevention (CDC) provides support for states to maintain registries that provide high-quality data.

Access the [National Program of Cancer Registries \(NPCR\)](#) website follow the steps below to learn more.

Step 1

Find the section **U.S. Cancer Statistics** and select the link for the **Data Visualizations Tool**. Explore the content provided; note that the dropdown menu at the top of the screen allow for different views. Consider the following questions:

- What cancer site had the highest number of cases for all races in 2017?
- What cancer site had the highest death rate for males in 2017?
- What cancer site had the highest incidence for females in 2017?
- What cancer site had the highest death rate for females in 2017?

Step 2

Return to the NPCR homepage. Find the section **Registry Tools and Software** and select each of the links to explore the following content:

- Collecting and processing registry data
- Editing registry data
- Securing registry data

Computerization in Health Informatics and Information Management

Whether disclosing patient information or using an encoder to help code a patient encounter, the systems used within HI help to maintain an efficient and productive environment for the proper management of patient health information. To further enhance your understanding of the topics presented in the lesson, read the following material.

Information System Trainer	Develops and implements the Information System (IS) training plan.
Systems Analyst	Performs tasks such as data collection, data analysis and creating data flow diagrams.
Mapping Specialist	Develops maps between different classifications and vocabularies. (i.e. ICD-9-CM and ICD-10-CM).
Data Analyst	Manages, interprets, analyzes health data into information that is accurate and consistent.
Chief Security Officer	Protects healthcare data from access, revision or destruction from unauthorized people.
Director of Clinical Informatics	Leads the implementation of an electronic health record system and then continues to oversee services post-implementation.

Discussion of EHRs

Implementation of the Electronic Health Record (EHR) has brought about many changes for health information (HI). As a result, many of the HI professional's core tasks have evolved to optimize efficiency and communication while focusing on the maintenance and management of patient health records. The digitizing of paper to

electronic has made information more accessible within and between healthcare organizations

Core Tasks

The HI professional uses computer software applications daily within the healthcare setting as part of the EHR and as part of the processing and managing of patient health information. **Software applications** are instructions that direct the computer to process data into meaningful information that is useful to the healthcare provider and users of the health information. Many of the HI professional's core tasks, including software applications, continuously evolve. The changing technologies affect record processing and the assembly process and transitioning from transcription to provider-created documentation. technology.

Administrative Information and Systems

Administrative information is used to manage the business of healthcare. This information is collected on each patient and includes such things as date of birth, insurance specifics, address, next of kin etc. This information is linked to the patient at the 'core' of the system and all clinical information connects as each encounter is added. Some of the key administrative systems used in healthcare organizations and provider practices include the following:

- Master patient index (MPI)
- Registration/patient access
- Scheduling
- Financials (i.e. patient accounts/billing and finance)

In some cases, gathering the different types of administrative information requires the use of separate systems that ultimately 'connect' to the organization's central information system. This ensures that the clinical and administrative information is linked so that each patient is provided with the correct information. When a patient enters the front door to start their healthcare journey, it is critical for healthcare professionals to accurately process this and each encounter that follows to allow for quality treatment and timely, accurate billing.

Quality Impact of the Master Patient Index

"The master patient index (MPI) provides the critical link among disparate health information systems. Its primary purpose is to facilitate access to clinical and financial information contained in those systems. This is true whether the index is a local MPI, a multi-entity enterprise MPI, or a regional health information exchange. The same solid, unchanging HIM principles of accuracy, completeness, and data integrity, important in creating a quality patient identification system, also serve as the foundation of the electronic health record and health information exchange"

Clinical Information Systems

Clinical Information is all the information used by providers and clinicians to make patient care decisions. This information is by and large collected and maintained in the healthcare organization's electronic health record (EHR) for each patient. *However, it is important to note that some organizations and practices still use either a paper-based health record or a hybrid*

version for their patients. The absence of an EHR is often due to the combination of cost, size of the organization, and volume of patients.

Many healthcare organizations do not use one central information system to document clinical information but rather may use different systems for some areas (i.e. Radiology, Laboratory). Therefore, the goal of a healthcare organization is to connect these separate systems via the EHR so that all clinical information is accessible. In turn, this requires HI professionals to frequently use, abstract, and maintain data within the Clinical Information Systems (CISs). These systems are central to the EHR, as their data is then used across all other systems to diagnose, treat, monitor, and manage patients. This data is tied to specific patients allowing providers and all healthcare clinicians to make accurate and informed decisions about patient healthcare.

The CISs allow for collection, storage, and retrieval of uniquely identified patient data. These systems are often specific to a particular area of treatment or functionality, such as the radiology information system (RIS) or the laboratory information system (LIS). The CIS is one part of a larger structure along with operations systems and administrative and finance systems. These various systems feed data into the EHR. HI professionals manage this information to ensure data integrity and that accreditation and regulatory standards are met.

Examples of clinical information systems include the following:

- Clinical Decision support
- Document Management System
- Radiology Information System
- Laboratory Information System
- Nursing Information System
- Pharmacy Information System
- Interdisciplinary Charting System

- Emergency Department System
- Anesthesia Information System
- Patient Monitoring System
- Telehealth
- Smart Cards
- Computerized Provider Order Entry (CPOE)
- Electronic Medication Administration Record (e-MAR)

Clinical decision support (CDS) involves a variety of tools used by clinicians and providers that make decisions regarding a patient's healthcare treatment. Examples of these tools include computerized alerts for medication interactions or allergies, reminders to providers, order-sets specific by diagnosis or procedure, and clinical guidelines. These tools can improve the efficiency, safety, quality, and effectiveness of patient healthcare by supporting consistency and accuracy of decision making while reducing errors and adverse events. To learn more about CDS, access HealthIT.gov.

Telehealth

Telehealth uses electronic devices, remote patient monitoring, and networks to support and provide patient care when the provider is at a different location, whether that be at a different hospital in a neighboring city, or somewhere across an ocean. Through telehealth care, rural patients can receive comparable treatment to those located in urban centers. Healthcare providers do not need to physically travel to the patient's location, and thus, lifesaving procedures can be performed faster. Nurses use telehealth to provide follow-up care and monitoring.

There are benefits and risks of using telehealth for the patient, provider, and healthcare organization. Study the two lists below comparing some benefits and risks for the patient. Then complete online research to identify some benefits and risks for the provider and healthcare organization.

Benefits

- Online patient portal
- Virtual appointments
- Personal health apps

Risks

- Gaps in medical care
- Overuse of medical care
- Unnecessary or overlapping medical care

Is the system **Administrative** or **Clinical**?

Master Patient
Index

Document
Management
System

Scheduling System

Laboratory
Information
System

Patient
Registration

Telehealth

Clinical

Administrative

Patient Health Information and the EHR

Healthcare Information (HI) professionals play a central role in managing the vast amount of health data and information that is collected within healthcare organizations. The electronic health record (EHR) is the primary tool used to manage patient longitudinal health history. To enhance patient care, it is important to use the EHR for tracking extensive data and converting it into meaningful information. However, the privacy and security of this data must be managed according to federal regulations to protect personal health information. HI professionals are experts in following appropriate processes and procedures while also bringing meaning to each patient story.

Components of the Electronic Health Record

The EHR has had a tremendous impact on the HI profession. It has transformed many of the paper-based functions that HI professionals typically performed including record assembly, analysis, coding, and release of information. Record assembly has been

eliminated as any loose paper documentation is scanned and 'filed' electronically. Some systems require authentication at the time of data entry so the need for analysis retrospectively is reduced if not totally eliminated. The changes brought on by the EHR have expanded the knowledge and skill set needed of the HI professional to managing the vast amount of data it contains, assuring data quality, managing projects, implementing, and maintaining information systems while still overseeing the privacy, security and release of the patient health information that was formally all on paper.

The privacy and security of this data must be managed according to federal regulations such as the Health Insurance Portability and Accountability Act (HIPAA) and the 2009 Health Information Technology for Economic and Clinical Health Act (HITECH), as well as applicable state law or accrediting agency guidelines.

Office of the National Coordinator for HIT

To assist with the transition to electronic health records, the U.S. government established the Office of the National Coordinator for Health Information Technology (ONC) in 2004. Organized under the Secretary of the U.S. Department of Health and Human Services, the ONC serves as "a resource to the entire health system to support the adoption of health information technology and the promotion of nationwide health information exchange to improve healthcare" (U.S. Department of Health and Human Services, About ONC).

Explore the following resources to help deepen your understanding of this government organization.

- Navigate the resources provided by HealthIT.gov
- Study the structure of the various ONC offices using the [ONC Organization Chart](#)
- Read about the [Benefits of EHRs](#)
- Read about the [Certification of Health IT](#)

Ensuring the Integrity of the EHR

"The goal of the electronic health record (EHR) is to help healthcare professionals produce and use quality data for evidence-based knowledge management and decision-making for patient care. The EHR has the potential to minimize medical errors if the data are accurate and meet quality criteria. EHRs are comprised of many different technologies. EHR governance by means of clear policies, standards, procedures, and functionalities should be established for all systems and applications that interface with the EHR, and define who owns and has responsibility for maintaining and creating the data dictionary for each system and module."